

AVERE: IMPROVING AUDIOVISUAL EMOTION REASONING WITH PREFERENCE OPTIMIZATION

Ashutosh Chaubey, Jiacheng Pang, Maksim Siniukov & Mohammad Soleymani

Institute for Creative Technologies

University of Southern California

Los Angeles, CA 90007, USA

achaubey@usc.edu & soleymani@ict.usc.edu

ABSTRACT

Emotion understanding is essential for building socially intelligent agents. Although recent multimodal large language models (MLLMs) have shown strong performance on this task, two key challenges remain: (i) spurious associations between emotions and irrelevant audiovisual cues (*reasoning errors*) and (ii) hallucination of audiovisual cues (*perception errors*) driven by text priors in the language model backbone. To quantify and understand these issues, we introduce **EmoRe-AIM**, a benchmark designed to evaluate MLLMs for cue–emotion associations, hallucinations and modality agreement. We then propose **AVEm-DPO**, a preference optimization technique that aligns model responses with both audiovisual inputs and emotion-centric queries. Specifically, we construct preferences over (i) responses exhibiting spurious associations or hallucinations and (ii) audiovisual input pairs guided by textual prompts. We also include a regularization term that penalizes reliance on text priors, thereby mitigating modality-specific cue hallucinations. Experimental results on DFEW, RAVDESS and EMER demonstrate that our method significantly improves the performance of the reference baseline models (6-19% of relative performance) in zero-shot settings. By providing both a rigorous benchmark and a robust optimization framework, this work enables principled evaluation and improvement of MLLMs for emotion understanding and social AI. Code, models and benchmark will be released at avere-iclr.github.io.

1 INTRODUCTION

Emotion understanding is essential for social AI agents to generate tailored responses and foster meaningful human–machine interactions (Chaturvedi et al., 2023; Kolomaznik et al., 2024; Elyoseph et al., 2024). Emotion perception also finds applications in domains such as health (Balcombe & De Leo, 2022; Litendahl et al., 2025) and education (Salloum et al., 2025), where appropriately responding to affective states can improve therapeutic alliance and learning outcomes.

Traditional multimodal emotion recognition methods (Sun et al., 2023; Wang et al., 2023; Chen et al., 2024) lack interpretability, as they only perform classification without grounding responses in audiovisual cues. Moreover, emotion is a complex and multi-componential construct that extends beyond the basic emotion labels that can be assigned by supervised learning methods (Ekman & Friesen, 1978; Scherer, 2005). To address these challenges, recent approaches leverage multimodal large language models (MLLMs) to generate detailed emotion descriptions for interpretability (Cheng et al., 2024; Huang et al., 2025a) and to output emotion-related keywords that cover a broader spectrum of emotional states (Lian et al., 2024; 2025a).

However, audiovisual MLLMs are susceptible to *hallucinations*, frequently generating inaccurate or fabricated responses (Li et al., 2023; Sahoo et al., 2024). In the context of emotion understanding, they face two critical bottlenecks, as illustrated in Fig. 1. First, these models often ground emotion predictions on irrelevant cues (e.g., attire color, ambient noise) – *reasoning errors*. Second, they tend to hallucinate additional cues in their responses to justify emotions – *perception errors*. Such hallucinations are largely driven by text priors in the language model backbone, which bias the model to include cues that commonly co-occur with specific emotions (e.g., associating tears

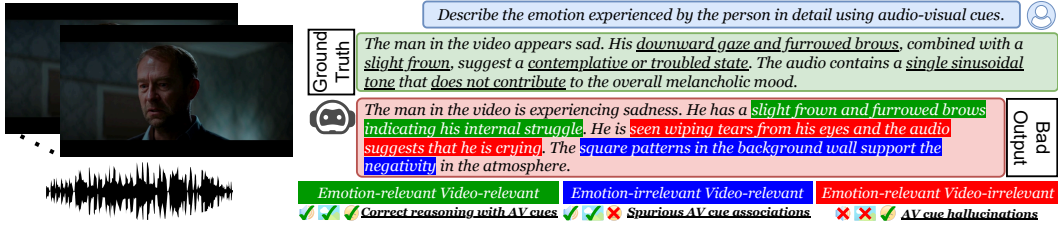


Figure 1: Existing MLLMs (i) include spurious associations between AV cues and emotions – *reasoning errors* (blue highlight) and (ii) hallucinate AV cues to explain emotions – *perception errors* (red highlight). AV: audiovisual.

with the sound of crying). The scarcity of high-quality, emotion-specific instruction tuning datasets (Cheng et al., 2024; Lian et al., 2025a) further aggravates these issues. Addressing these challenges is essential, as they compromise the reliability of MLLM agents in social interactions and complex emotion reasoning scenarios.

Existing emotion reasoning benchmarks (Lian et al., 2023b; 2024) lack the diverse and complex samples needed to fully evaluate these issues. Additionally, current audiovisual hallucination benchmarks (Sung-Bin et al., 2025; Leng et al., 2025) predominantly focus on object-level hallucinations in audio or video, rather than on emotion-specific reasoning. Moreover, many existing MLLMs (Cheng et al., 2024; Lian et al., 2025a) rely on two-stage evaluation pipelines involving an external (often proprietary) LLM such as GPT (OpenAI et al., 2024), making replication and benchmarking difficult. To address these limitations, we introduce the **EmoReAIM** benchmark, a comprehensive suite of multiple-choice question–answer (MCQA) tasks designed to evaluate audiovisual emotion reasoning, modality agreement and hallucination-related stress tests (Fig. 2). The MCQA format enables transparent, reproducible and scalable evaluation of MLLMs on emotion-centric tasks without requiring additional LLMs during inference.

Evaluation of recent MLLMs on our benchmark highlights spurious association and hallucination issues outlined in Fig. 1. To address these limitations, we propose **AVEm-DPO** – a multimodal direct preference optimization (DPO) technique (Rafailov et al., 2023) to enhance the emotion reasoning capabilities of MLLMs. In particular, we design explicit prompt-based audiovisual input preferences to mitigate hallucinations caused by cross-modal interactions. We also introduce text-prior debiasing, which penalizes policy reward for responses to text-only inputs. Together, these techniques significantly improve the performance of reference MLLMs, outperforming all baselines in zero-shot evaluation on both our benchmark and existing emotion recognition and reasoning datasets.

To summarize, the main contributions of our work are:

- We introduce the **EmoReAIM** benchmark with **4000 human-verified** MCQA samples to evaluate emotion reasoning and emotion-related hallucinations in MLLMs, highlighting bottlenecks such as spurious audiovisual cue associations and hallucinated cues for explaining emotions.
- We propose **AVEm-DPO**, a direct preference optimization technique that enforces explicit prompt-based modality preferences and reduces text-only model biases through a regularizer that penalizes over-reliance on text priors.
- We conduct extensive evaluations of existing MLLMs, demonstrating current bottlenecks and showing the superior performance of the proposed DPO-trained models in zero-shot settings.

2 RELATED WORK

MLLMs for Emotion. While general MLLMs (Zhang et al., 2024; Lin et al., 2024; Zhang et al., 2025a; Xu et al., 2025b; Li & team, 2025) show non-trivial emotion recognition ability (Cheng et al., 2024), several studies pursue domain-specific instruction tuning (Xie et al., 2024; Chaubey et al., 2025; Yang et al., 2025). EmotionLLaMA (Cheng et al., 2024) is an audiovisual LLM for emotion recognition and captioning, finetuned on a limited dataset ($\approx 30k$ samples). Lian et al. (2024) introduces open-vocabulary emotion recognition (OV-ER), and AffectGPT (Lian et al., 2025a) employs a lightweight audiovisual fusion projector for OV-ER. EmotionQwen (Huang et al., 2025a) improves emotion understanding while preserving general skills via a mixture-of-experts router. Han

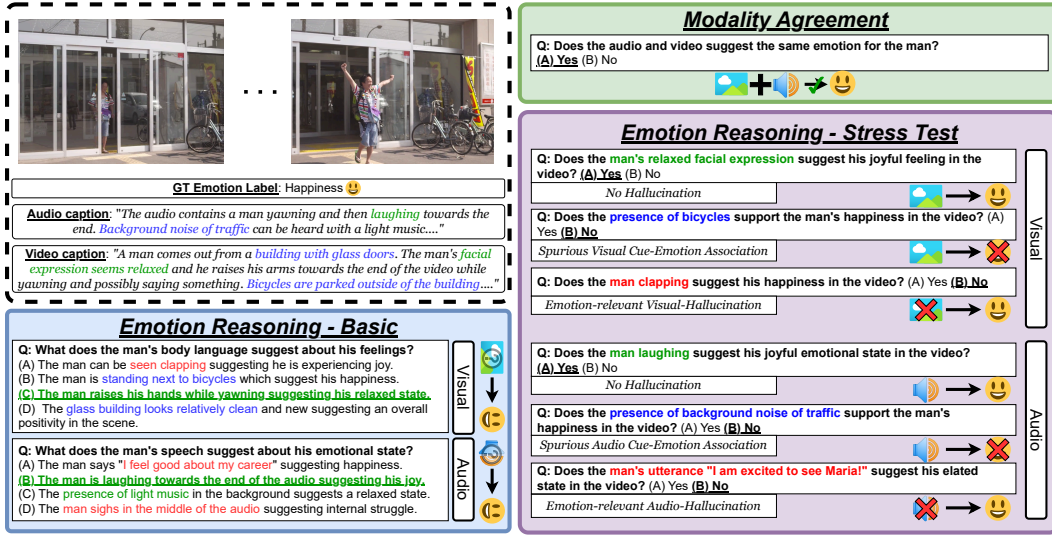


Figure 2: **EmoReAIM Tasks**. In addition to basic emotion reasoning, we include tasks for *Modality Agreement* and *Emotion Reasoning - Stress Test* to test spurious cue-emotion associations and cue hallucinations. **Red text** is a hallucinated cue, **blue text** is an emotion-irrelevant cue and **green text** is a cue relevant for emotion understanding. Correct choices are underlined.

et al. (2025b) use modality-specific experts with attention reallocation to handle audiovisual emotion mismatch, and Wen et al. (2025) leverage retrieval-augmented generation with chain-of-thought for better reasoning. In contrast, we improve reasoning through multimodal preference optimization and text-prior debiasing.

Rigorous evaluation of multimodal emotion reasoning requires diverse, systematic benchmarks. Lian et al. (2023b) provide detailed descriptions of transcript, audio and visual cues for emotion reasoning, which can support GPT-based evaluation (Cheng et al., 2024; Han et al., 2025b). Xing et al. (2025) present a holistic benchmark spanning text, image, video and audio hallucinations related to emotions. Our benchmark instead focuses squarely on audiovisual emotion understanding with a standardized pipeline and tasks beyond hallucination, including modality agreement and spurious cue–emotion associations.

Preference Optimization. Direct preference optimization (DPO) (Rafailov et al., 2023; Liu et al., 2025a) was introduced to align LLMs to human preferences. DPO has also emerged as a leading approach for mitigating hallucinations in vision LLMs (Yu et al., 2024; Wang et al., 2024; Sarkar et al., 2025; Huang et al., 2025b; Liu et al., 2025b; Zhang et al., 2025b), but its use in audiovisual LLMs remains limited. VistaDPO (Huang et al., 2025b) increases video LLM robustness by building instance-level, temporal-level and object-level preferences of video inputs. Sun et al. (2025) apply process DPO for step-wise audiovisual reasoning, while Tang et al. (2025) use multi-round DPO for audiovisual captioning. Luo et al. (2025) employ DPO for emotional speech alignment to improve Omni-LLM outputs. Ye et al. (2025) construct multimodal preference data via ambiguity scoring, and Lian (2025) use group relative policy optimization for AffectGPT. Concurrently, Omni-DPO (Chen et al., 2025) studies audiovisual modality preference. Our method differs by constructing prompt-based audiovisual preference pairs for fine-grained alignment and by introducing text-prior debiasing to reduce hallucinations in MLLMs.

3 EMOREALM BENCHMARK

Fig. 2 shows different tasks present in the proposed **EmoReAIM** Benchmark. The goal of this benchmark is to test the reasoning capabilities of MLLMs to judge the *emotion experienced by the character in the given video*, specifically over the following verticals – (i) **reasoning the correct emotion** with relevant audiovisual cues (ii) identifying whether the inferred emotion from **audio and video are in agreement** (iii) testing the **association of perceived audiovisual cues** with different

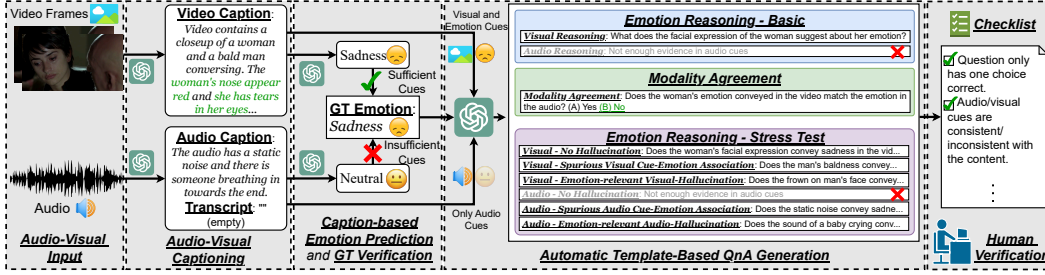


Figure 3: **EmoReAIM Creation Pipeline.** We first disentangle the audiovisual information by separate captioning and verify the cues with text-based emotion prediction to find emotion-relevant cues. Finally, GPT-4o is used to generate MCQA samples that are later verified manually.

emotions (*reasoning errors*) and (iv) testing **audiovisual hallucination due to text-only emotion-related biases** (*perception errors*).

3.1 TASK DESCRIPTIONS

Emotion Reasoning – Basic. This task evaluates an MLLM’s ability to identify and reason about the emotion experienced by a person in a video by linking appropriate audio (e.g., speech transcription, tone) and visual (e.g., facial expression, body language) cues to specific emotions. To increase difficulty, the ground-truth emotion is not provided in the question. Incorrect options are constructed by modifying the correct answer to include either **emotion-irrelevant cues present in the video** or **hallucinated cues** that falsely justify the emotion.

Modality Agreement. This task assesses whether the audio and visual modalities convey the same emotional state. Unlike AVHBench (Sung-Bin et al., 2025), which focuses on general cross-modal alignment, this task specifically targets agreement in emotional interpretation across modalities.

Emotion Reasoning – Stress Test. MLLMs are vulnerable to both *reasoning errors* and *perception errors*: the former lead the model to base its responses on irrelevant audiovisual cues present in the input, while the latter cause it to rely on hallucinated cues that are not actually present. This task probes MLLMs for susceptibility to spurious cue-emotion associations (*perception errors*) and hallucinated explanations driven by language model biases (*reasoning errors*). Each question follows the format: “Does the {audio/visual cue} suggest {emotion} of the character?”. For a modality X, we define three sub-tasks: (i) No Hallucination — correctly associating an audio/visual cue with the appropriate emotion. (ii) Spurious X Cue-Emotion Association — linking emotion-irrelevant cues to the correct emotion. (iii) Emotion-Relevant X-Hallucination — associating the correct emotion with a hallucinated cue that typically co-occurs with it. For example, in Fig. 2, a man is not clapping (per the visual caption), yet a hallucination-based question associates clapping with happiness—since clapping is commonly linked to positive emotions like joy.

3.2 AUTOMATIC DATA CREATION

Fig. 3 shows the automatic pipeline used to construct the *EmoReAIM* benchmark. Our approach builds on existing manually labeled audiovisual emotion recognition datasets that provide single-word emotion annotations. For each video, we first use an MLLM to extract detailed audio and visual captions separately, effectively disentangling the two modalities. These captions describe both emotion-relevant and irrelevant cues. To verify whether either modality reflects an emotion, we prompt an LLM to classify the audio and video captions independently into one of seven categories of neutral, in addition to six basic emotions Ekman (2005). Samples are discarded if neither caption yields a valid emotion label. Given the validated captions and emotion label, we then generate tailored prompts and question templates for each task described in Section 3.1. This modality-wise captioning and emotion verification process ensures the construction of high-quality, verifiable MCQA pairs that reflect meaningful audiovisual cue associations. More details and prompts are present in Section B.

Details. All videos are sourced from the DFEW dataset (Jiang et al., 2020). GPT-4o (OpenAI et al., 2024) is used for caption extraction, emotion classification and question–answer pair generation.

3.3 POST-PROCESSING AND HUMAN VERIFICATION

We employ GPT-4o (OpenAI et al., 2024), Gemini-2.5 (Gemini-Team et al., 2025) and Qwen-2.5 (Qwen-Team et al., 2025) to predict the correct answer to the generated questions just by using question text as input. We remove all the QA pairs for which all the models identified the correct answer just with the text information. Finally, since the QA samples are generated automatically leveraging MLLMs, which can hallucinate themselves, we perform a human verification over the samples generated by recruiting over 470 participants using the crowd-sourcing platform Prolific. Details are present in Section B.2.

3.4 BENCHMARK STATISTICS

Table 1 summarizes the data statistics of the proposed *EmoReAIM* Benchmark, which comprises a total of **4,000 questions** over **2,649 unique videos**. Samples from the benchmark are present in Section B.5. Importantly, for tasks which always have a fixed set of answer choices (*Emotion Reasoning - Stress Test* and *Modality Agreement - Yes/No*), we ensure that there is a uniform distribution of correct answer texts over the possible answer choice texts. Additionally, we ensure that the distribution of emotion labels over the videos in the benchmark matches the video source dataset (refer to Section B.3 for details). It is also important to note that *EmoReAIM* is only used as a **test set** to evaluate the reasoning capabilities of MLLMs, and we use a different dataset for preference optimization (refer Section 4.3).

Table 1: *EmoReAIM* Benchmark Statistics.

Task		# QA	# vid.	Rand. Acc.
Reasoning Basic	Audio	972	784	25%
	Visual	1024	883	25%
Modality Agreement	Audio	456	456	50%
	Visual	820	655	50%
Reas. Stress Test	Audio	728	593	50%
	Visual	728	593	50%
Total		4000	2649	

4 AVEM-DPO

Direct preference optimization (DPO) (Rafailov et al., 2023) aligns LLMs to human preferences, bypassing the need to develop a reward model. In the context of audiovisual LLMs, given a reference model π_{ref} , we can reformulate the DPO objective to learn an optimal policy π_{θ} as the following,

$$\max_{\pi_{\theta}} \mathbb{E}_{(a,v,x) \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot | a,v,x)} [r(a,v,x,y)] - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta}(\cdot | a,v,x) \parallel \pi_{\text{ref}}(\cdot | a,v,x)) \quad (1)$$

where (a,v) is audiovisual input, x is text prompt, y is text response and $r(a,v,x,y)$ is the reward function for given input-output pair. Optimizing Eq. (1) to find optimal policy results in the following reward formulation,

$$r(a,v,x,y) = \beta \log \frac{\pi_{\theta}(y | a,v,x)}{\pi_{\text{ref}}(y | a,v,x)} + \beta \log Z(a,v,x) \quad (2)$$

where $Z(\cdot)$ is the partition function derived in Rafailov et al. (2023). With access to a preference dataset $\mathcal{D}_y^{\text{pref}}$ with samples (a,v,x,y_w,y_l) and using the Bradley-Terry preference model (Bradley & Terry, 1952) to model preference of chosen response (y_w) over rejected response (y_l), the final DPO objective becomes

$$\mathcal{L}_{\text{DPO}} = -\mathbb{E}_{(a,v,x,y_w,y_l) \sim \mathcal{D}^{\text{pref}}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | a,v,x)}{\pi_{\text{ref}}(y_w | a,v,x)} - \beta \log \frac{\pi_{\theta}(y_l | a,v,x)}{\pi_{\text{ref}}(y_l | a,v,x)} \right) \right] \quad (3)$$

4.1 MULTIMODAL PREFERENCE OPTIMIZATION

Naive DPO (Eq. (3)) applied to MLLMs, when relying only on response preference, often causes the policy model to overfit to the input prompt x while neglecting the multimodal inputs during alignment (Wang et al., 2024; Sarkar et al., 2025). To address this limitation, preference optimization can be extended to incorporate audiovisual inputs as follows:

$$\mathcal{L}_{\text{DPO}}^{av} = -\mathbb{E} \left[\log \sigma(u(a_w, v_w, a_l, v_l, x, y_w)) \right], \quad u(\cdot) = \beta \log \frac{\pi_{\theta}(y_w | a_w, v_w, x)}{\pi_{\text{ref}}(y_w | a_w, v_w, x)} - \beta \log \frac{\pi_{\theta}(y_l | a_l, v_l, x)}{\pi_{\text{ref}}(y_l | a_l, v_l, x)} \quad (4)$$

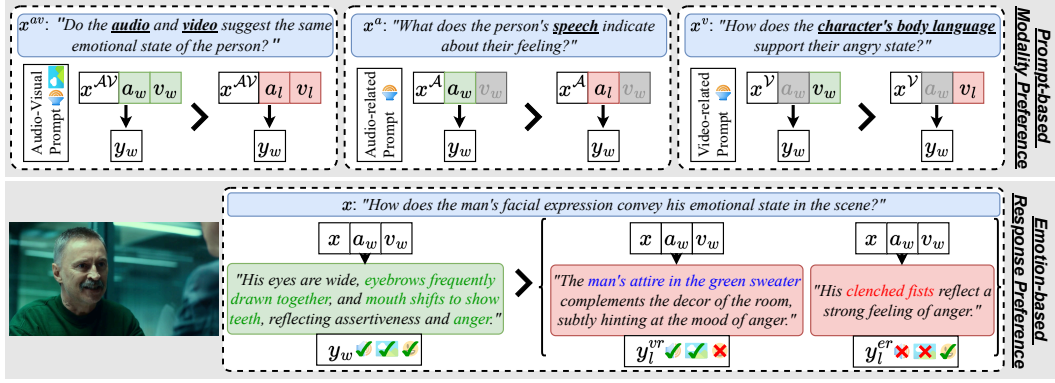


Figure 4: **Preference pairs in AVEm-DPO.** (Top) Fine-grained preference over modality input based on current prompt. (Bottom) Each chosen response y_w has two rejected responses – y_l^{vr} relevant to the video but with spurious emotion association and y_l^{er} irrelevant to the video (hallucinated) but related to the emotion.

where (a_w, v_w) and (a_l, v_l) denote the chosen and rejected multimodal inputs. This objective ensures that the policy model aligns its response y_w to the correct (chosen) audiovisual input (a_w, v_w) .

Prompt-based Modality Preference (PMP). While Eq. (4) enforces preference over *non-text* inputs, in the case of audiovisual (or “*omni*”) LLMs the input prompt x^m may relate to both audio and visual modalities, or to only one of them ($m \in \mathcal{M} = \{\mathcal{AV}, \mathcal{A}, \mathcal{V}\}$). This often leads to cross-modality-induced hallucinations in MLLMs (Sung-Bin et al., 2025), where a response to a prompt concerning one modality x^{m_1} is spuriously influenced by another modality $m_2 \in \mathcal{M} \setminus \{m_1\}$.

To mitigate this issue, we construct the preference dataset $\mathcal{D}_{av}^{\text{pref}}$ with fine-grained modality-level preferences conditioned on the input prompt x^m , as illustrated in Fig. 4 (Top). For example, for a query specific to one modality x^m (e.g., visual: “How does the character’s body language support their angry state?”), we modify only the corresponding input(s) of modality m (i.e. visual) in the rejected pair, thereby enforcing that the model’s response remains grounded in that modality. Thus, our prompt-based modality preference objective becomes,

$$\mathcal{L}_{\text{DPO}}^{av-\text{prompt}} = -\mathbb{E}[\log \sigma(u(a_w, v_w, a_l^{\text{PMP}}, v_l^{\text{PMP}}, x^m, y_w))] \quad (5)$$

where $a_l^{\text{PMP}} = a_w$, iff $m = \mathcal{V}$ and $v_l^{\text{PMP}} = v_w$, iff $m = \mathcal{A}$. We perform multiple forms of negative sampling for constructing (a_l, v_l) (see Section 5.2); however, because our task is emotion reasoning, the best results were achieved when we choose the rejected audiovisual input to be a sample with an emotion different from the chosen input (a_w, v_w) .

Emotion-based Response Preference. To mitigate spurious cue-emotion associations and hallucinations described in Section 1, for a given input (a_w, v_w, x) we construct two rejected responses that are variations of the chosen response y_w , as illustrated in Fig. 4 (Bottom). Specifically, y_l^{vr} includes an audio/visual cue that is relevant to the audiovisual input but does not explain the emotion, whereas y_l^{er} introduces audio/visual cues related to the emotion but absent from the audiovisual input (hallucinated). Following Huang et al. (2025b), we assign weights to these rejected responses in the DPO loss in Eq. (3) as,

$$\mathcal{L}_{\text{DPO}}^y = -\mathbb{E}_{(a_w, v_w, x, y_w, y_l^{vr}, y_l^{er}) \sim \mathcal{D}_y^{\text{pref}}} \left[\log \sigma \left[\beta \left(\log \frac{\pi_\theta(y_w | a_w, v_w, x)}{\pi_{\text{ref}}(y_w | a_w, v_w, x)} - \sum_{i \in \{vr, er\}} \beta_i \log \frac{\pi_\theta(y_l^i | a_w, v_w, x)}{\pi_{\text{ref}}(y_l^i | a_w, v_w, x)} \right) \right] \right] \quad (6)$$

where $\beta_{er} + \beta_{vr} = 1$. This formulation establishes strong contrasts between chosen and rejected responses, encouraging the policy model to ground its outputs in correct and emotion-relevant audiovisual cues. Unlike Huang et al. (2025b), however, we do not include completely irrelevant responses as rejections in DPO based on empirical findings in Section E.6.

4.2 TEXT PRIOR DEBIASING (TPD)

Audiovisual LLMs have strong text priors that cause them to hallucinate and include cues in their response, which usually occur together (e.g., the presence of a crying person accompanied by the

sound of crying). To suppress such behaviour, we propose to penalize the reward $r(a, v, x, y)$ derived in Eq. (2) to generate the response using only text input as follows,

$$r(a, v, x, y) = \beta \log \frac{\pi_\theta(y | a, v, x)}{\pi_{\text{ref}}(y | a, v, x)} + \beta \log Z(a, v, x) - \gamma_{\text{TPD}} \log \pi_{\text{text}}(y | x) \quad (7)$$

where π_{text} is a trained language model and γ_{TPD} is a hyperparameter. In our experiments, we choose π_{text} to be the language model backbone in π_{ref} . This penalty ensures that the responses that are explainable purely by text priors get discounted and responses supported by audio/video get relative credit. Plugging Eq. (7) in the Bradley Terry model results in the following objective,

$$\mathcal{L}_{\text{DPO-TPD}} = -\mathbb{E}_{(a, v, x, y_w, y_l) \sim \mathcal{D}^{\text{pref}}} \left[\log \sigma \left(\beta \left(\log \frac{\pi_\theta(y_w | (a, v, x))}{\pi_{\text{ref}}(y_w | (a, v, x))} - \log \frac{\pi_\theta(y_l | (a, v, x))}{\pi_{\text{ref}}(y_l | (a, v, x))} \right) - \gamma_{\text{TPD}} \left(\log \pi_{\text{text}}(y_w | x) - \log \pi_{\text{text}}(y_l | x) \right) \right) \right] \quad (8)$$

where (a, v) denote (a_w, v_w) for simplicity. During training, we stop gradients through π_{text} as it is just used to identify the text priors that a language model has. To maintain the text-only capabilities of the language model backbone, we attach LoRA module (Hu et al., 2022) to it for training. To accommodate two rejected responses, we perform scaling similar to Eq. (6) on the rejected responses in the TPD term as described in Section C.1 (Eq. (8)) to get the final TPD objective $\mathcal{L}_{\text{DPO-TPD}}^y$. The final objective function of **AVEm-DPO** is as follows,

$$\mathcal{L}_{\text{AVEm-DPO}} = \mathcal{L}_{\text{DPO-TPD}}^y + \lambda_{av} \mathcal{L}_{\text{DPO}}^{av-prompt} \quad (9)$$

where λ_{av} is a hyperparameter. Implementation details are present in Section C.3.

4.3 PREFERENCE DATA

For AVEm-DPO training, we construct preference data using a pipeline similar to Fig. 3. This preference dataset is different from *EmoReAIM*, which we exclusively use for testing. We use MAFW (Liu et al., 2022) and a subset of MER2025 (Lian et al., 2025b) *Track-1 train set* as the source datasets to create preference samples. We prompt Gemini-2.5 Gemini-Team et al. (2025) to generate variations of the correct answers (chosen responses) to the questions where the audiovisual cue is altered to be either a spurious emotion-related video-relevant cue (y_l^{vr}) or a hallucinated cue related to the emotion present (y_l^{er}). Note that we do not perform any manual verification on the generated data, which still results in a performance gain demonstrating the efficiency of the proposed approach. Details in Section C.2.

5 EXPERIMENTS

Datasets & Metrics. For *EmoReAIM* benchmark, we report the average accuracy per task for all the tasks. For tasks with *Yes/No* responses, we additionally report the precision, recall and F1 score following previous multimodal hallucination benchmarks (Sung-Bin et al., 2025; Li et al., 2023). Beyond *EmoReAIM*, we also evaluate on established emotion recognition datasets—DFEW (Jiang et al., 2020), RAVDESS (Livingstone & Russo, 2018), MER2023 (Lian et al., 2023a)—and the emotion reasoning dataset EMER (Lian et al., 2023b). None of these datasets is used in training to ensure zero-shot evaluation. Following prior work (Cheng et al., 2024; Han et al., 2025b), we report unweighted and weighted average recalls for DFEW and RAVDESS and weighted F1 for MER2023. For emotion reasoning, we adopt GPT-based evaluation (Cheng et al., 2024), comparing generated responses against ground truth. In addition to clue and label overlap, we assess two dimensions: (i) *spurious cue-emotion associations*, where irrelevant cues are linked to emotions, and (ii) *hallucinatory cues*, where non-existent audiovisual cues are fabricated. For all metrics, higher values indicate better performance. Further details are provided in Section D.1.

Reference models. We use two audiovisual MLLMs as reference – EmotionLLaMA (Cheng et al., 2024) and our own developed base model. Our model is similar to EmotionLLaMA in architecture with changes to the audio encoder (*whisper-large-v3* (Radford et al., 2023)) and video encoder (*LanguageBind* (Zhu et al., 2024)). For EmotionLLaMA, we remove the text (subtitle) input branch to be consistent with the other baselines and retrain the model on the original dataset without subtitles – denoted as **EmotionLLaMA*** (Cheng et al., 2024). More details in Section D.2.

Table 2: Zero-shot performance comparison of different methods on existing audiovisual emotion recognition benchmarks. Mod. are the modalities input to the model with the prompt. A: Audio, V: Video, T: Text Subtitles. ‡: evaluation without text subtitle input.

Model	Mod.	DFEW		RAVDESS		MER2023 F1	EMER			
		UAR	WAR	UAR	WAR		Clue	Label	Spurious	Halluc.
VideoLLaMA 2	A,V	43.65	48.66	41.81	31.62	50.79	3.82	3.80	4.25	4.23
OLA	A,V	38.17	41.73	27.45	22.11	55.82	3.80	3.33	3.93	4.22
VITA-1.5	A,V	39.31	42.56	50.67	46.88	66.94	4.77	4.72	5.16	5.70
Qwen-2.5 Omni	A,V	46.94	54.34	32.88	28.05	79.72	5.85	6.78	6.39	6.21
EmotionLLaMA	A,V,T	45.59	59.37	28.20	29.24	90.36	6.03	6.99	5.89	5.26
EmotionLLaMA [‡]	A,V	42.72	54.06	30.36	30.45	89.05	2.76	2.78	3.44	2.36
MoSEAR	A,V,T	44.48	56.60	-	-	90.27	-	-	-	-
Our base	A,V	56.78	60.14	53.59	53.01	89.19	5.63	6.45	5.41	5.19
+ Naive-DPO		55.67	59.90	53.63	52.94	88.59	5.81	6.30	5.96	5.48
+ Vista-DPO [†]		56.42	62.33	56.94	53.64	90.06	6.08	6.89	6.58	6.07
+ AVEm-DPO		58.54	64.24	58.66	55.48	92.18	6.37	7.08	7.09	6.75
EmotionLLaMA*	A,V	54.89	58.26	52.59	48.12	90.01	5.78	6.21	5.36	5.23
+ Naive-DPO		54.97	58.12	52.69	49.01	89.35	5.89	6.35	5.89	5.62
+ Vista-DPO [†]		56.28	61.58	56.42	50.96	91.19	6.05	6.56	6.85	6.31
+ AVEm-DPO		57.06	62.12	56.21	51.03	91.68	6.02	6.99	7.02	6.62

Baseline Preference Optimization Approaches. We compare with original **Naive-DPO** (Rafailov et al., 2023) using single rejected samples from our DPO data and modified Vista-DPO (Huang et al., 2025b) for audiovisual inputs – denoted as **Vista-DPO[†]** (Section D.3 for details).

5.1 EMOTION REASONING AND RECOGNITION RESULTS

EmoReAIM Results. Table 3 presents the performance of different approaches on the proposed *EmoReAIM* benchmark. AVEm-DPO achieves substantial gains over the reference models, demonstrating the effectiveness of multimodal preference optimization and text-prior debiasing. While the baselines perform strongly on basic reasoning tasks, Table 3 shows that they struggle on *Modality Agreement* and *Stress-Test* evaluations (Expanded table in Section E.1 and Table 13).

Notably, our preference optimization also surpasses Vista-DPO and Naive-DPO by significant margins. To further examine the bottlenecks in baseline models, Section E.1 reports results on samples probing spurious audiovisual-emotion correlations and hallucinated cues. For state-of-the-art systems such as Qwen 2.5 Omni (Xu et al., 2025b) and VITA-1.5 Fu et al. (2025), hallucination emerges as a more severe issue than spurious cue-emotion associations. Moreover, unlike findings from Sung-Bin et al. (2025), our results indicate that audio and visual hallucinations are equally prevalent in emotion reasoning tasks. Additionally, Table 13 shows the performance of video-only and audio-only baselines and reveals that multimodal inputs hurt reasoning capabilities.

Emotion Recognition and Reasoning on Existing Benchmarks. Table 2 (expanded in Section E.3) shows the performance on existing emotion benchmarks mentioned before. We can notice that our reference models outperform baselines, showing the efficacy of reference in understanding emotion. Moreover, preference tuning additionally boosts the performance, especially for emotion reasoning on

Table 3: Performance comparison of different methods on the proposed *EmoReAIM* Benchmark.

Model	Reas. Basic		Modality Agree. F1	Reas. - Stress	
	Audio Acc.	Visual Acc.		Audio F1	Visual F1
VideoLLaMA2	63.1	66.8	52.5	53.2	58.4
OLA	63.2	60.4	42.7	56.6	54.8
VITA-1.5	63.1	84.3	30.2	52.8	56.3
Qwen 2.5 Omni	76.8	89.2	33.3	55.0	56.8
Our base	69.2	85.3	34.6	50.3	59.9
+ Naive-DPO	71.3	85.9	41.6	54.8	65.9
+ Vista-DPO [†]	72.4	87.8	52.1	73.6	86.7
+ AVEm-DPO	77.9	92.5	60.0	80.9	94.6
Emot.-LLaMA*	64.8	84.9	33.1	46.7	63.2
+ Naive-DPO	67.2	85.7	42.8	52.6	67.6
+ Vista-DPO [†]	69.0	86.9	40.9	68.6	87.3
+ AVEm-DPO	76.5	89.9	56.8	75.4	91.7

Table 4: User evaluation on EMER.

Model	Emot.↑	Assoc.↑	Incons.↓
VideoLLaMA 2	9.82%	0.75%	15.38%
OLA	9.36%	7.46%	5.58%
VITA 1.5	11.60%	17.25%	6.04%
Qwen 2.5 Omni	10.75%	18.57%	10.13%
EmotionLLaMA	1.89%	11.53%	68.61%
Our + AVEm-DPO	54.74%	43.35%	4.67%

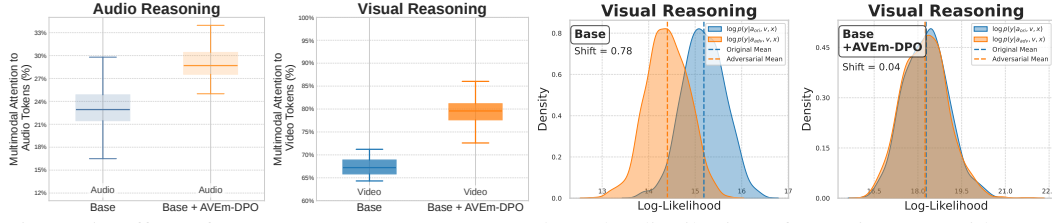


Figure 5: Effect of AVEm-DPO on – (Left two plots) the distribution of attention over video and audio tokens taken as a percentage over the total attention over all multimodal tokens for audio and visual reasoning tasks in *EmoReAlM*; (Right two plots) the log-likelihood distribution shift of the correct answer for visual reasoning tasks on corrupting the audio input a_{ori} with adversary a_{adv} .

EMER, reducing spurious cue-emotion associations and hallucinations. It is important to note that previous emotion MLLM baselines (Cheng et al., 2024; Han et al., 2025b) use text subtitle as additional input. Qualitative comparison to baselines is present in Section F. While most baselines perform poorly on the out-of-domain RAVDESS dataset, our reference and preference-tuned models perform significantly better, showing their generalizability.

User evaluation. We perform a user evaluation with 40 participants on EMER generations from different models and report results in Table 4. Participants chose our model the most for emotion description and emotion-cue associations and the least for inconsistencies. (Details in Section E.4).

5.2 ANALYSIS

Ablation Study. Table 5 shows the performance of the preference-tuned model after removing the proposed components of AVEm-DPO. We perform this analysis on *EmoReAlM* and report the average metrics over audio and visual reasoning (Section D.5 for details). Removal of any of the key components results in a significant performance drop, especially for the reasoning tasks. Moreover, ablating TPD results in a huge performance drop on the hallucination stress test samples, underlining its efficacy in eliminating cue hallucinations in audio-visual emotion reasoning.

Table 5: Ablation study over different components of the proposed AVEm-DPO approach. PMP: Prompt-based Modality Preference, ERP: Emotion-based Response Preference, TPD: Text Prior Debiasing.

Method	Basic.	Agree.	Stress	Spur.	Hall.
Our base	77.3	34.6	55.1	47.3	39.2
+ AVEm-DPO	85.2	60.1	87.8	92.7	97.6
w/o PMP	81.0	54.9	79.6	86.2	88.1
w/o ERP	81.8	56.2	79.4	84.9	88.4
w/o TPD	83.8	58.9	78.8	87.1	77.8
+ Contr. Dec.	79.1	51.3	61.7	50.9	54.8

Comparison with training-free contrastive decoding. Similar to VCD Leng et al. (2024), we perform contrastive decoding using diffused audiovisual inputs and report results in Table 5 (last row), showcasing it is significantly worse than AVEm-DPO.

Design Choices and Sensitivity to Hyperparams. Section E.5 shows that prompt-based modality preference using a different emotion audiovisual (AV) input as (a_l, v_l) works better compared to using random videos or diffused versions of the inputs. Section E.6 shows that using emotion-relevant and video-relevant rejected responses (y_l^{er}, y_l^{vr}) works better compared to only using one or using a completely irrelevant response. Section E.7 detail the sensitivity of AVEm-DPO to various hyperparameters, highlighting the role of various components in eliminating spurious cue-emotion associations and hallucinations.

Attention redistribution after AVEm-DPO. To analyze the effect of preference optimization on model attention, we plot the distribution of aggregate multimodal input attention over audio and visual tokens averaged over all attention heads for audio and visual reasoning tasks in *EmoReAlM* in Fig. 5 (left two plots). We can observe that the attention over relevant modality increases after AVEm-DPO, ensuring consistent model responses grounded on the relevant modality. More attention redistribution experiments are present in Section E.8.

Robustness to adversarial inputs. As shown in Fig. 12 (Section E.9), the model response on a prompt relevant to one modality should not change on changing the input of the irrelevant modality.

To test this robustness on visual reasoning tasks, we plot the distribution of log-likelihoods of correct responses for our base and AVEm-DPO models and show the distribution shift using Kernel Density Estimation (KDE) on changing the audio input in Fig. 5(right two plots). AVEm-DPO trained model results in negligible shifts, showing its robustness. Detailed analysis in Section E.9.

5.3 VALIDITY OF GENERATED PREFERENCE DATA

As mentioned in Section 4.3, our preference dataset is automatically generated using Gemini 2.5 (Gemini-Team et al., 2025). Performing human verification on the entire training data is too costly. Therefore, to show the validity of the generated preference tuning data, we perform human verification on a subset of 1000 random samples from the generated data with the help of 90 participants recruited through Prolific (Prolific). Each generated sample is verified by three or more annotators. As shown in Table 6, for the different categories of preference responses mentioned in Section 4.1 – chosen (y_w), video-relevant rejected (y_l^{vr}), and emotion-relevant rejected (y_l^{er}) – we report the number of samples in which the majority of annotators found the generated responses correct. These results validate our automatically generated preference data.

Table 6: Human verification statistics on generated preference data.

Response type	# Total verified	# Majority correct	# One or more correct
Chosen (y_w)	1000	912	967
Rejected - Video Relevant (y_l^{vr})	1000	895	923
Rejected - Emotion Relevant (y_l^{er})	1000	856	912

6 LIMITATIONS AND FUTURE WORK

The proposed EmoReAIM benchmark is derived from the DFEW (Jiang et al., 2020) dataset, leveraging its emotion labels, and hence, it may inherit its cultural biases. Additionally, since our benchmark and training data are derived from existing emotion recognition datasets with short videos (~ 2 -10 seconds), long video emotion understanding and reasoning remain an open topic that can be addressed in future work.

Although the proposed AVEm-DPO significantly improves the reference model’s performance, a few limitations remain. Similar to other baselines, our model trained with AVEm-DPO performs poorly on the recognition for *disgust* (an ambiguous emotion (Hendel et al., 2023)) as shown in Section E.3 and Table 15. We attribute this to the limited amount of training samples available for this emotion class. Moreover, a closer look at the performance on the subtasks of the *Emotion Reasoning - Stress Test* task of EmoReAIM (Section E.2 and Table 14) reveals that there is still room for improvement to mitigate spurious audio cue-emotion associations.

7 CONCLUSION

This work addresses the bottlenecks of emotion reasoning in MLLMs, with two major contributions – *EmoReAIM* Benchmark for evaluating emotion reasoning over a complex and diverse set of tasks and *AVEm-DPO* preference optimization technique to mitigate bottlenecks of MLLMs such as spurious audiovisual cue-emotion associations and audiovisual cue hallucinations. The proposed method outperforms open-source baselines on the proposed and existing emotion understanding benchmarks under a zero-shot setting. Moreover, a detailed ablation study with analysis of attention redistribution and log-likelihood shift upon preference tuning supports the efficacy of the proposed prompt-based modality preference and text-prior debiasing approaches.

ETHICS STATEMENT

This work builds upon publicly available audiovisual datasets for research purposes, specifically DFEW for benchmark creation (Section 3.2) and MAFW/MER2025 for preference optimization (Section 4.3). We did not collect new audiovisual data, ensuring no additional privacy risks. All data usage complies with the licensing terms of the original datasets. To mitigate potential harms, the released *EmoReAIM* benchmark will only contain automatically generated and human-verified question-answer pairs; users must independently obtain the underlying videos from the original sources under appropriate licenses. For human verification (Section 3.3) and user studies (Table 4),

participants were recruited via Prolific and compensated at fair rates commensurate with task requirements and participant location, aligning with ethical standards for crowd work. We ensured informed consent, anonymity and the right to withdraw at any point. The proposed methods aim to improve reliability in emotion reasoning by reducing hallucinations and spurious cue associations in multimodal large language models. However, emotion recognition and inference from audiovisual data can carry risks of misinterpretation, bias reinforcement, or misuse in surveillance and high-stakes applications. Moreover, users of the proposed method are advised to read the limitations of the proposed approach mentioned in Section 6 to avoid potential safety concerns. We emphasize that our benchmark and models are intended strictly for academic research, with the goal of advancing robust, interpretable and socially responsible AI. We caution against deployment in sensitive real-world contexts (e.g., healthcare, hiring, law enforcement) without careful domain-specific validation and safeguards.

REPRODUCIBILITY STATEMENT

To ensure reproducibility and transparency, we provide additional details about data creation and experiments in the Appendix. All the prompts used for data creation are present in Section B.1. Implementation details for the proposed method, along with hyperparameter settings, are provided in Sections C.3 and D.2, while the details about the baseline approaches are present in Sections D.3 and D.4. Details about human verification of the benchmark and user evaluation are present in Sections B.2 and E.4. Evaluation metrics are detailed in Section D.1. We also provide the detailed setup for our ablations in Section D.5. Our benchmark, code and model weights will be made publicly available upon acceptance to ensure reproducibility and ease of use for the proposed work. Code, models and benchmark will be released at [avere-iclr.github.io](https://github.com/avere-iclr).

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